STAFF REVIEW · H446 055 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Two different keys map to the same bucket.

2. When a reference to the insertion node is already known.

3. Ordered search can require O(log n) comparisons.

4. 2, 8, 4. The queue supplies 4, 8, 2; the stack reverses that order.

I Do — source worked steps

1. Explain the trace — Stack (LIFO): the most recently visited earlier page is retrieved first. Browser navigation may use multiple stacks for back/forward.

We Do — guided CHECK

1. Explain the trace
Queue FIFO serves earliest arrivals first, subject to real-world priority policies.

You Do — source mark schemes

1. Stack [1], Queue [1], Dictionary/hash table [1], Graph [1].

2. 2D array room×period [2]; indexed direct access [2]; fixed rectangular dimensions suit [1]; graph unnecessary for this access [1].

3. List random indexing O(1) [2]; LinkedList random index traversal O(n) [1]; List middle insertion O(n) shifts [1]; linked list pointer insertion with known node O(1) [2].

Exit Ticket — source answer

1. False. Choose from operations, sizes and constraints.